6.05 Motion in a Circle			
6.05a 6.05b 6.05c	Uniform motion in a circle	a) Understand and be able to use the definitions of angular velocity, velocity, speed and acceleration in relation to a particle moving in a circular path, or a point rotating in a circle, with constant speed. *Includes the use of both ω and $\dot{\theta}$.* b) Be able to use and apply the relationships $v = r\dot{\theta}$ and $a = \frac{v^2}{r} = r\dot{\theta}^2 = v\dot{\theta}$ for motion in a circle with constant speed. c) Be able to solve problems regarding motion in a horizontal circle. *e.g.* *Motion of a conical pendulum.* *Motion on a banked track.* *Problems will be restricted to those involving constant forces but learners will be required to resolve forces in two dimensions.*	
6.05d 6.05e	Motion in a vertical circle	d) Understand the motion of a particle in a circle with variable speed. *In 'Stage 1' Learners will be expected to use energy considerations to calculate the speed of a particle at a given point on a circular path.*	e) Extend their understanding of the motion of a particle in a circle with variable speed to include the radial and tangential components of the acceleration.

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
6.05f			f) Be able to solve problems involving motion round a vertical circle including motion which is not restricted to a circular path. *This is restricted to a combination of motion in a circle and free fall.* *e.g. The subsequent motion of a particle moving on the outside of a smooth circular surface.* *The motion of a particle on a string moving in a vertical circle and then as a projectile.*